Voice Scribe v4 — The Stitch Policy, by Intuition

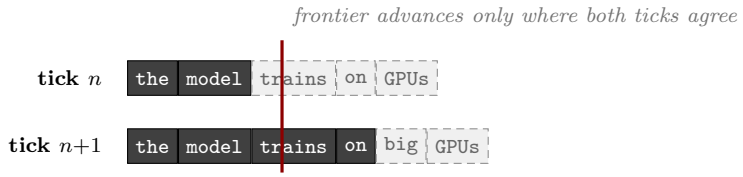
Every tuning parameter of `stream_policy.py`: the failure without it, and what the tuned value buys

2026-08-04 (policy as of commit `9876d50`)

Legend. `word` committed (permanent) `word` tentative (may still change) `word` wrong text `word` duplicate | commit frontier

0. The engine

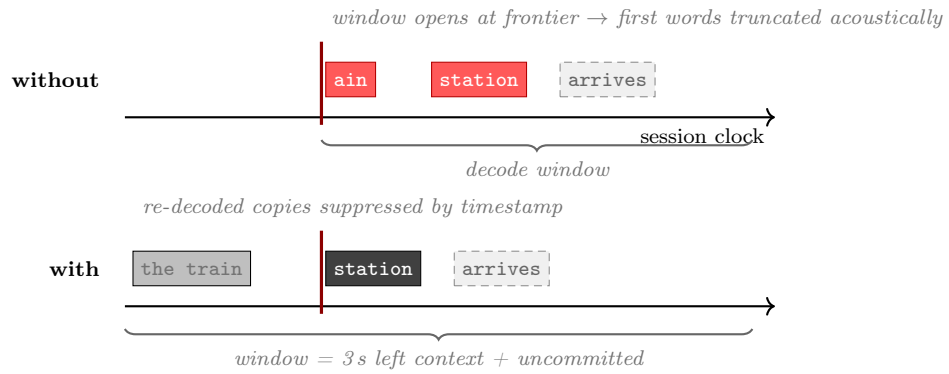
Every 750 ms tick the server re-decodes one window: 3 s of already-committed context plus all uncommitted audio. The policy compares this hypothesis with the previous tick's and commits the longest *agreeing prefix* (case- and punctuation-insensitive). Everything else stays tentative. Structural rule (C5.5): every word either commits or withholds — *none revises*. That is why "zero retractions" is a property of the construction, not a hope.



The hypotheses agreed on `trains on`; those commit. The tail (`big GPUs` vs `GPUs`) disagrees; it waits. Ten knobs bound this loop. Each figure below shows the degenerate setting first — the failure the knob exists to prevent — then the tuned behavior.

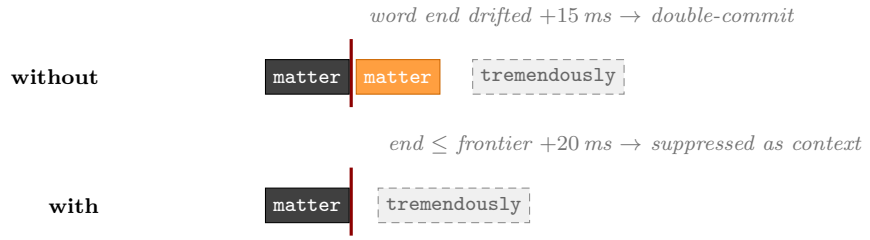
1. `left_context_s = 3.0` — the decoder never starts cold

Without: the window opens exactly at the frontier; the model decodes mid-phrase with no conditioning and mangles the first words. **With:** 3 s of committed audio is re-decoded for conditioning only; `localize()` drops the copies *by timestamp*, never by string match (string matching fails exactly when the model re-punctuates).



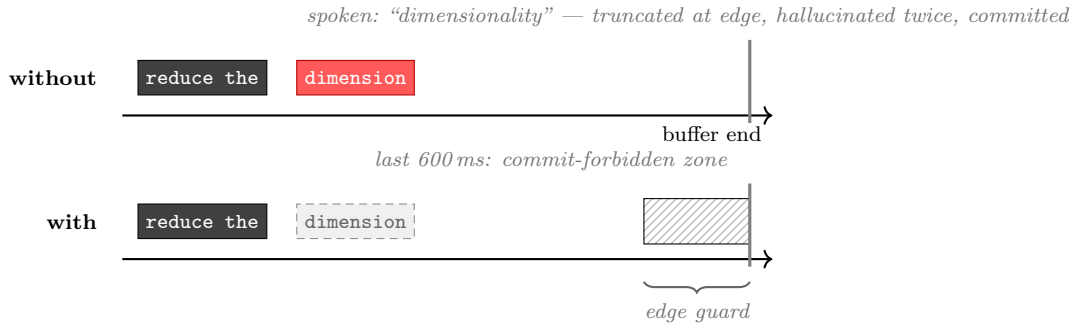
2. `frame_tolerance_s = 0.02` — one decoder frame of drift

Without: a re-decode re-times the last committed word's end 15 ms past the frontier; it survives the boundary filter and commits *again* — append-only means the duplicate is permanent and invisible to the retraction reducer. **With:** anything ending within one Whisper frame (20 ms) of the frontier is treated as re-decoded committed audio.



3. `edge_guard_s = 0.6` — agreement is not enough at the edge

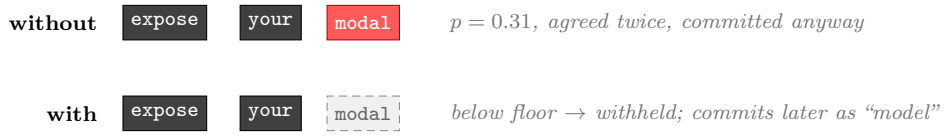
Without: a word cut off at the buffer edge makes the model hallucinate a completion — *the same completion, with good probability, two ticks in a row.* Agreement commits it; the real word later differs. **With:** words ending in the last 600 ms can never commit, agreed or not. This is the structural defense (C7.4): distance from the edge, not confidence, is the test.



4. `word_prob_floor = 0.5` — a veto, not a rejection

Without: an agreed word at 0.31 probability commits permanently. **With:** the commit scan stops at the first agreed word under the floor; the word stays tentative and re-tries next tick with more

audio behind it. Vetoes only withhold — they never edit — so the retraction guarantee holds.



5. The terminal-word rule (structural — no knob)

The last word of a hypothesis never commits: Whisper re-punctuates and re-capitalizes it on nearly every pass, and committed text takes the surface form of the *newer* hypothesis. Field note (2026-08-04 wire log): this is exactly why sentence-final punctuation drops when a commit boundary lands on a sentence end — the period existed only while the word was terminal.



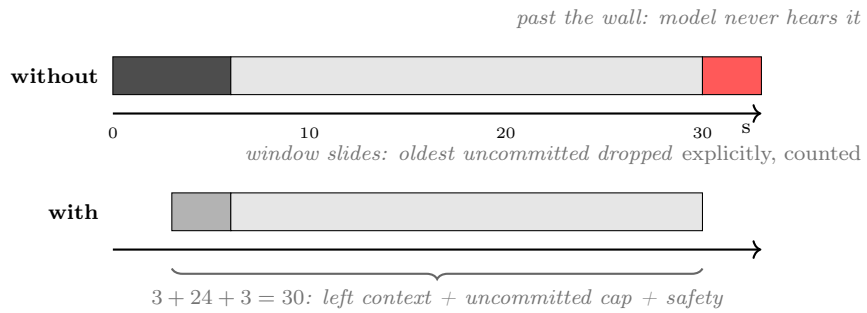
6. `max_holdback_ticks = 6` — the anti-stall valve

Without: disfluent speech makes consecutive hypotheses flip-flop; the agreed prefix stops growing and the frontier stalls *forever* — unbounded commit lag. **With:** after 6 stalled ticks ($6 \times 750\,\text{ms} = 4.5\,\text{s}$) the newest hypothesis force-commits without agreement or the probability floor — but never past the edge guard. The valve's ledger from the 2026-08-04 field run (impromptu speech): `force_commit_rate` 8.6% — how often it fired; `forced_span_wer` 14.3% — the quality price of committing unagreed text; `commit_lag_max` 4.1–6.1 s — the bound at work. This trade is the primary v4.2 tuning surface.



7. `max_uncommitted_s = 24` + `safety_s = 3` — the 30 s wall

Whisper's input is a fixed 30 s window; `pad_or_trim` *silently truncates* anything longer. The invariant $3 + 24 + 3 = 30$ (enforced in `validate()`) makes silent truncation impossible. **Without:** the uncommitted span grows past the wall and the model simply never hears the overflow — unaccounted audio loss. **With:** `plan_window` reports the overflow and the session drops the *oldest* uncommitted samples explicitly, counted in `dropped_audio_ms`. The 2026-08-04 u3 run slid three 79 s utterances through this and reported `dropped_audio_ms = 0` with zero retractions — the slide worked without ever needing the drop.

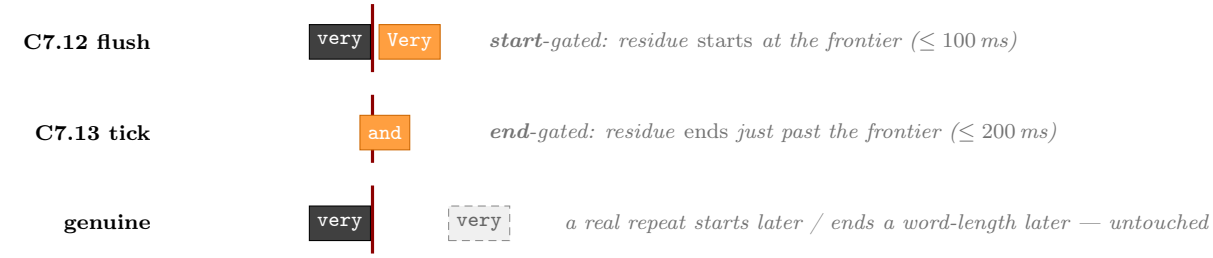


8. `flush_left_context_s = 1.0` — context priced in latency

The flush decode (end of utterance — silence is ground truth, commit everything) opens only 1 s before the frontier, not 3: the flush sits on the finalization-latency path, and *every second of context here is paid directly in the user-visible finalization number* (gate: ≤ 0.5 s after the first utterance; 2026-08-04 measured 0.544 s). `validate()` refuses a flush context larger than `left_context_s` — retention only guarantees that much audio still exists.

9. The twin residue guards — `seam_tolerance_s = 0.10` (C7.12) and `tick_residue_tolerance_s = 0.20` (C7.13)

Same disease, two symptoms. The frontier is a DTW word-end *estimate* that lands tens of ms early, so acoustic residue of the last committed word leaks into the next decode and re-emits ("very Very", 2026-08-02 field finding). Both guards are conjunctions on the *leading* word matching the last committed word — but they gate on opposite ends, because the impostors differ:



Accepted residual risk, stated in the code: a genuine immediate repeat falling inside the tolerance is indistinguishable by any local rule and is dropped — so every suppression is counted on the session-close log line, keeping the trade observable. 2026-08-04 field ledger: 3 seam catches, 2–14 tick-residue suppressions per session, zero visible duplicates.

Coda — how the knobs met the field on 2026-08-04

One recording of disfluent, self-correcting impromptu speech exercised every mechanism above in a single afternoon: the holdback valve fired at 8.6% (its ledger, §6), the slide handled 3×79 s utterances without dropping a sample (§7), the twin guards caught every frontier residue (§9), the terminal-word rule explained the missing sentence periods (§5), and the flush context sat 44 ms over its latency gate (§8). Tuning any one knob moves a number you can now name.